Sahaj Bhadja

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Summary

Software Engineer pursuing an M.S. in Computer Science at USC with 2+ years building **cloud-native products**, **log-driven troubleshooting workflows**, and reliable **data/operations pipelines** with strong cross-functional collaboration across engineering, QA, and operations teams.

Education

University of Southern California **Los Angeles, CA**
Master of Science in Computer Science *August 2024 – May 2026*
GPA: 3.75/4 Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University **Gandhinagar, India**
B.Tech. in Computer Science and Engineering *November 2020 – May 2024*
GPA: 3.88/4 Coursework: OOP, Software Engineering, Operating Systems, Data Retrieval

Experience

USC Dornsife **LA, California**
Software Engineer *September 2025 – Present*

- Built **Python** workflow services and data interfaces for **500+ users**, reducing latency from **1.2s to 350ms** through query optimization while improving backend observability and production debugging in **cloud-based** systems.
- Implemented fault-tolerant orchestration for **5K+ monthly jobs** with validation safeguards and rollback checks, improving **issue triage speed**, data reliability, and handoff quality between engineering and operational support teams.

Grad8 **LA, California**
Software Engineer Intern *June 2025 – August 2025*

- Developed **React** and **Flask** features handling **10K+ daily requests**, helping triage incoming bugs from telemetry and user reports, then delivering fixes with clear documentation for QA and product stakeholders.
- Improved release velocity with **CI/CD** controls and concurrent processing, reducing turnaround time by **60%** while preserving quality through automated checks, root-cause analysis, and concise issue-resolution writeups.

USC AI Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*

- Built **Python** experimentation pipelines over **100K+ trajectory logs**, identifying behavior anomalies and failure patterns through structured data analysis that improved reproducibility and accelerated debugging across model evaluation workflows.
- Produced technical summaries and diagnostics dashboards that converted raw findings into actionable engineering tasks, improving **cross-team communication** and reducing ambiguity during issue triage and implementation planning cycles.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*

- Engineered **API** and **ETL** microservice workflows with OOP, AWS Glue, and S3, improving response times by **30%** while strengthening operational data flows and reliability across distributed business-critical services.
- Maintained **20+** unit and integration suites using Jenkins, PyTest, and Mockito across **Linux/Windows** environments, boosting pipeline efficiency by **68%** and reducing technical debt by **20%** via automation.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*

- Developed **REST API** and **WebSocket** modules supporting eight workflows and **2M+ synchronized records**, improving system stability while coordinating incident details and fixes across operations and engineering stakeholders.
- Automated deployments and testing with **Docker** and **Jenkins**, reducing deployment time by **50%** and improving release consistency through script-driven validation, clear rollback plans, and disciplined runbook documentation.

Skills

Languages: Python, JavaScript, TypeScript, Java, C++, SQL, Bash
Backend & APIs: Flask, FastAPI, REST APIs, WebSockets, Microservices, ETL Pipelines
Cloud & Infrastructure: AWS (S3, Glue, ECS), Azure, Docker, Kubernetes, Linux/UNIX
Observability & Ops: Log Analysis, System Metrics, Issue Triage, Root-Cause Analysis, Documentation
Engineering Practices: Unit/Integration Testing, CI/CD, Git, Jenkins, GitHub Actions, Agile/Scrum

Projects

Linguistix | *Python, AWS, Docker, CI/CD*

- Developed and deployed a containerized **RESTful API** on AWS CodePipeline and ECS/Fargate with automated **CI/CD** and validation checks, achieving **87% accuracy** and production-grade reliability under sustained operational workloads.

HMIS | *MERN, ABDM APIs, Twilio, Sentry, TypeScript*

- Built a healthcare workflow platform integrating **ABDM APIs** and **Sentry** monitoring, improving operational visibility and reducing administrative load by **30%** through actionable issue tracking and faster resolution loops.

Chat2SQL | *Java, TypeScript, SQL, Docker, Maven*

- Built an AI-assisted SQL workspace supporting **10+ databases**, improving query authoring and schema exploration through structured data access patterns, repeatable validations, and robust multi-environment troubleshooting workflows.